

IP Addressing (part 3)

Binary Conversion Exercises

In last week's post, I explained how binary works under the hood of every IP address (the place value table, what bits mean, and why computers speak in 0s and 1s).

This week, we will be practicing with some exercises, so this gets stuck to memory. I will show every single step so that even if you are hearing about binary conversion for the very first time, you can follow along from start to finish.

PART 1: Converting Decimal to Binary

We will use the **Division by 2** method. Here are the steps:

- Take your number and divide it by 2
- Write down the quotient and the remainder
- Keep dividing the quotient by 2 until you reach 0
- Read all the remainders from **Bottom** to **Top**
- The sequence of 0s and 1s is your binary number
- If you get less than 8 digits, add zeros to the **left** until you have exactly 8 digits.

Every IP address has 4 sections called **octets**. You convert each octet separately

Example 1: 192.168.1.1 → Binary

Octet 1: 192

Division	Quotient	Remainder
$192 \div 2$	96	0

$96 \div 2$	48	0
$48 \div 2$	24	0
$24 \div 2$	12	0
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Read remainders BOTTOM to TOP:

11000000

Octet 2: 168

Division	Quotient	Remainder
$168 \div 2$	84	0
$84 \div 2$	42	0
$42 \div 2$	21	0
$21 \div 2$	10	1
$10 \div 2$	5	0
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

Read remainders BOTTOM to TOP:

10101000

Octet 3: 1

Division	Quotient	Remainder
$1 \div 2$	0	1

Read remainders **BOTTOM to TOP**: 00000001

Only 1 digit from the division. Pad with 7 zeros on the left to make 8 bits: 00000001

Octet 4: 1

Same calculation as Octet 3 above: 00000001

Final Binary Answer	11000000.10101000.00000001.00000001
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Note:

Class	C	The first octet is 192. The rule says any first octet between 192 and 223 is Class C. 192 sits right at the start of that range.
Network Portion	192.168.1	Class C uses the first 3 octets to identify the network. Think of it like your street name. Everyone on the same network shares these 3 numbers.
Host Portion	.1	The last octet identifies the specific device. Think of it like your house number. This device is house number 1 on that street.

Public / Private	PRIVATE	This starts with 192.168 which is one of only three private address ranges in the world. Private means it only works inside a local network like your house or office. It cannot travel on the open internet. In fact, 192.168.1.1 is the default login address for most home routers in Nigeria and worldwide.
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Example 2: 10.0.0.5 → Binary

Octet 1: 10

Division	Quotient	Remainder
$10 \div 2$	5	0
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

Read remainders BOTTOM to TOP: 00001010

Only 4 digits. Pad with zeros on the left: 00001010

Octet 2: 0

0 in binary is always all zeros: 00000000

Octet 3: 0

Same as in octet 2 above: 00000000

Octet 4: 5

Division	Quotient	Remainder
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$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

Read remainders **BOTTOM to TOP**: **00000101**

Only 3 digits. Pad with zeros on the left: 00000101

Final Binary Answer	00001010.00000000.00000000.00000101
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Note:

Class	A	The first octet is 10. Any first octet between 1 and 126 is Class A. 10 falls right inside that range.
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Network Portion	10	Class A only uses the very first octet to identify the network. Just that one number. This leaves 3 full octets free for devices which means massive space.
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Host Portion	.0.0.5	The last 3 octets all belong to the host in Class A. This gives the network room for over 16 million unique device addresses.
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Public / Private	PRIVATE	Any address where the first octet is exactly 10 is automatically private Class A. It does not matter what the remaining 3 octets are. The moment you see 10.anything.anything.anything, it is private. This is why big enterprises and cloud platforms like Azure use it. They need millions of addresses.
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PART 2: Converting Binary Back to Decimal

This direction uses the **place value table**. Each of the 8-bit positions has a fixed value that never changes:

Position 7	Position 6	Position 5	Position 4	Position 3	Position 2	Position 1	Position 0
128	64	32	16	8	4	2	1

- Write out the binary number and line it up with the place values above
- Wherever there is a 1, write down that position's value
- Wherever there is a 0, ignore it
- Add up all the values you wrote down because the total is your decimal number
- Do this for all 4 octets then put them together with dots

Example 1: 11001100.00001010.00000001.00000010 → Decimal

Octet 1: 11001100

128	64	32	16	8	4	2	1
1	1	0	0	1	1	0	0
128	64	-	-	8	4	-	-
Add active values: 128 + 64 + 8 + 4							204

Octet 2: 00001010

128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0
-	-	-	-	8	-	2	-
Add active values: 8 + 2							10

Octet 3: 00000001

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	1
-	-	-	-	-	-	-	1
Add active values: 1							1

Octet 4: 00000010

128	64	32	16	8	4	2	1
0	0	0	0	0	0	1	0
-	-	-	-	-	-	2	-
Add active values: 2							2

Final Decimal Answer	204.10.1.2
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Note:

Class	C	First octet is 204. That falls between 192 and 223, which is the Class C range.
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Network Portion	204.10.1	Class C uses the first 3 octets for the network.
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Host Portion	.2	The last octet identifies the host device.
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Public / Private	PUBLIC	Run it through all three private checks. Does it start with 10? No. Does it start with 172 with second octet 16 to 31? No. Does it start with 192.168? No. It fails all three tests so it is public. This address lives on the open internet and can be reached from anywhere in the world.
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Example 2: 10000000.00000001.00000100.00001111 → Decimal

Octet 1: 10000000

128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0
128	-	-	-	-	-	-	-
Add active values: 128							128

Octet 2: 00000001

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	1
-	-	-	-	-	-	-	1

Add active values: 1

1

Octet 3: 00000100

128	64	32	16	8	4	2	1
0	0	0	0	0	1	0	0
-	-	-	-	-	4	-	-

Add active values: 4

4

Octet 4: 00001111

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1
-	-	-	-	8	4	2	1

Add active values: $8 + 4 + 2 + 1 = 15$

15

Final Decimal Answer

128.1.4.15

Note:

Class	B	First octet is 128. That sits right at the very start of the Class B range which runs from 128 to 191.
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Network Portion	128.1	Class B uses the first 2 octets for the network.
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Host Portion	.4.15	The last 2 octets identify the host device.
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Public / Private	PUBLIC	People often assume all 172 addresses are private but that is not true. The private B range only covers 172.16 to 172.31. This address starts with 128 not 172 so it was never going to be private. It is a fully public Class B address that lives on the open internet.
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One thing that clicked for me doing these exercises is that binary conversion looks scary until you actually do it step by step.